Sina Darabi

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Bio

- Sina Darabi is a Senior Research Engineer at the Barcelona Supercomputing Center (BSC-CNS), Spain. His research focuses on computer architecture and high-performance computing (HPC) for AI, with particular emphasis on networking, memory systems, and high-throughput processor design. He develops hardware–software co-design techniques to improve scalability, energy efficiency, and performance predictability for next-generation AI and datacenter systems. His work has been published in leading computer architecture and systems/networking venues, including ISCA, MICRO, SIGMETRICS, and NSDI.

Education

2015 – 2022
- **Ph.D., Sharif University of Technology** in Computer Engineering.
Major: *Computer Systems Architecture*
Advisor: *Professor Hamid Sarbazi-Azad*
Thesis: *Improve memory resource utilization in GPU by exploiting heterogeneity in applications, (Best Ph.D thesis Award)*

2013 – 2015
- **M.S., Isfahan University of Technology** in Computer Engineering.
Major: *Computer Systems Architecture*
Advisors: *Professor Masoudreza Hashemi* and *Professor Hossein Saidi*
Thesis: *Packet classification based on GPU architecture*

2007 – 2012
- **B.S., Isfahan University of Technology** in Computer Engineering.
Major: *Hardware Engineering*
Advisor: *Professor Shadrokh Samavi*
Thesis: *An ASIC based hardware implementation for image edge detection*

Employment History

2026–present
- **Senior Research Engineer.** Barcelona Supercomputing Center (BSC-CNS), Barcelona, Spain.

2023–2026
- **Postdoctoral Researcher.** Università della Svizzera italiana (USI), Lugano, Switzerland.

2022–2023
- **Postdoctoral Researcher.** Institute for Research in Fundamental Sciences (IPM), Tehran, Iran.

Awards and Achievements

2024
- Passed the first stage of the SNSF Ambizione grant evaluation.

2023
- Received the best PhD thesis award.
"Winner of the best computer science Ph.D. thesis in Iran chosen by the Computer Society of Iran, significant impact in their field."

2022
- The first PhD student in Computer Architecture with three top-tier conference publications from Iranian universities
- Published in Micro 2022 conference as the first paper from Iranian universities.
- 1350$ grant award for attending SIGMETRICS 2022

Awards and Achievements (continued)

2012 30000$ grant for developing pacemaker by Kermanshah University of Medical Sciences

Publications

Conference Publications

1. F. Babaei, M. Dolati, M. Mozhganfar, **S. Darabi**, and F. Tashtarian, "Resource management for distributed binary neural networks in programmable data plane," in *Proceedings of the IEEE/IFIP Network Operations and Management Symposium (NOMS)*, To appear, 2026.
2. **S. Darabi** et al., "Lazy-llc: Reducing gpu llc thrashing with shared-data and reusability filters," in *Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026)*, **To appear. Top-tier conference in computer architecture**, 2026.
3. M. Hosseini, **S. Darabi**, H. B. Pasandi, P. Eugster, and M. Choopani, "Cachecatalyst: Enhancing web caching for the latency-constrained internet," in *Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 2026)*, **Top-tier conference in networking and systems.**, 2026.
4. M. Hosseini, **S. Darabi**, H. B. Pasandi, M. Nakhjiri, and P. Eugster, "Application-driven reexamination of datacenter microbursts," in *Abstracts of the 2025 ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems*, **Top-tier conference in computer systems**, 2025, pp. 28–30.
5. D. Rovelli, C. Faerber, G. McKenzie, A. Pahlevan, **S. Darabi**, and P. Eugster, "Nano-consensus: Ultra-fast, quorum-less coordination on the wire," in *Proceedings of the 2025 ACM Symposium on Cloud Computing*, 2025, pp. 659–672.
6. N. Akbarzadeh, **S. Darabi**, A. Gheibi-Fetrat, A. Mirzaei, M. Sadrosadati, and H. Sarbazi-Azad, "A high-bandwidth high-capacity hybrid 3d memory for gpus," in *Abstracts of the 2024 ACM SIGMETRICS/IFIP PERFORMANCE Joint International Conference on Measurement and Modeling of Computer Systems*, **Top-tier conference in computer systems**, 2024, pp. 67–68.
7. M. Hosseini, **S. Darabi**, M. Nakhjiri, and P. Eugster, "Uncovering secrets of microbursts in datacenter network traffic," in *2024 20th International Conference on Network and Service Management (CNSM)*, IEEE, 2024, pp. 1–5.
8. **S. Darabi**, N. Mahani, H. Baxishi, E. Yousefzadeh, M. Sadrosadati, and H. Sarbazi-Azad, "Nura: A framework for supporting non-uniform resource accesses in gpus," in *Abstract Proceedings of the 2022 ACM SIGMETRICS/IFIP PERFORMANCE Joint International Conference on Measurement and Modeling of Computer Systems*, **Top-tier conference in computer systems**, 2022, pp. 39–40.
9. **S. Darabi** et al., "Morpheus: Extending the last level cache capacity in gpu systems using idle gpu core resources," in *2022 55th IEEE/ACM International Symposium on Microarchitecture (MICRO)*, **Top-tier conference in computer architecture**, IEEE, 2022, pp. 228–244.
10. S. M. Hosseini, A. H. Jahangir, and **S. Darabi**, "Session-persistent load balancing for clustered web servers without acting as a reverse-proxy," in *2021 17th International Conference on Network and Service Management (CNSM)*, IEEE, 2021, pp. 360–364.
11. N. Rohbani, **S. Darabi**, and H. Sarbazi-Azad, "Pf-dram: A precharge-free dram structure," in *2021 ACM/IEEE 48th Annual International Symposium on Computer Architecture (ISCA)*, **Top-tier conference in computer architecture**, IEEE, 2021, pp. 126–138.
12. Y. Darabimoghadam and **S. Darabi**, "A simple method for producing a pacemaker with low cost that works in modes: Voo, aoo," in *2011 2nd International Conference on Engineering and Industries (ICEI)*, IEEE, 2011, pp. 1–4.

Journal Publications

1. P. Peykani Sani, A. Mohammadpur-Fard, S. Mangipudi, **S. Darabi**, F. Reggazzoni, and P. Eugster, “Fast inter-enclave communication encryption,” *IEEE Computer Architecture Letters*, 2026.
2. M. Hosseini, **S. Darabi**, H. B. Pasandi, M. Nakhjiri, and P. Eugster, “Poison comes in small packages: Application-driven reexamination of datacenter microbursts,” *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 9, no. 2, pp. 1–23, 2025.
3. N. Mahani et al., “A low-latency on-chip cache hierarchy for load-to-use stall reduction in gpus,” *ACM Transactions on Architecture and Code Optimization*, 2025.
4. N. Akbarzadeh, **S. Darabi**, A. Gheibi-Fetrat, A. Mirzaei, M. Sadrosadati, and H. Sarbazi-Azad, “H3dm: A high-bandwidth high-capacity hybrid 3d memory design for gpus,” *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 8, no. 1, pp. 1–28, 2024.
5. A. Mohammadpur-Fard, **S. Darabi**, H. Falahati, N. Mahani, and H. Sarbazi-Azad, “Exploiting direct memory operands in gpu instructions,” *IEEE Computer Architecture Letters*, 2024.
6. M. Hosseini, **S. Darabi**, A. H. Jahangir, and A. Movaghar, “Yuz: Improving performance of cluster-based services by near-l4 session-persistent load balancing,” *IEEE Transactions on Network and Service Management*, 2023.
7. **S. Darabi**, N. Mahani, H. Baxishi, E. Yousefzadeh-Asl-Miandoab, M. Sadrosadati, and H. Sarbazi-Azad, “Nura: A framework for supporting non-uniform resource accesses in gpus,” *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 6, no. 1, pp. 1–27, 2022.
8. **S. Darabi** et al., “Osm: Off-chip shared memory for gpus,” *IEEE Transactions on Parallel and Distributed Systems*, vol. 33, no. 12, pp. 3415–3429, 2022.

Workshop Publications

1. H. B. Pasandi, M. Hosseini, **S. Darabi**, F. Rousseau, and J. A. Fraire, “Adaptive orbital direct-to-device: Rethinking satellite iot architecture,” in *Proceedings of the 20th Workshop on Mobility in the Evolving Internet Architecture*, 2025, pp. 38–43.
2. H. B. Pasandi, F. Rousseau, T. Nadeem, and **S. Darabi**, “Beyond terrestrial: Llm-orchestrated dts-iot networks for ubiquitous edge intelligence,” in *Proceedings of the 2025 ACM Workshop on Access Networks with Artificial Intelligence*, 2025, pp. 41–45.
3. H. B. Pasandi, Z. Tan, M. Hosseini, **S. Darabi**, F. Rousseau, and J. A. Fraire, “Fedsat-lam: Enabling large ai models on resource-constrained satellites via hierarchical federated learning,” in *Proceedings of the 2nd International Workshop on Edge and Mobile Foundation Models*, 2025, pp. 19–24.
4. A. Sadr, K. Pakzad, M. Hosseini, H. B. Pasandi, and **S. Darabi**, “Toward efficient layer-4 load balancing: A hybrid stateful–stateless approach,” in *Proceedings of the 2025 Conference on Network Systems and Architectures*, 2025.
5. M. Hosseini, **S. Darabi**, P. Eugster, M. Choopani, and A. H. Jahangir, “Rethinking web caching: An optimization for the latency-constrained internet,” in *Proceedings of the 23rd ACM Workshop on Hot Topics in Networks*, **The best networking workshop**, 2024, pp. 326–334.

Patents

1. N. Rohbani, S. Darabi-Moghaddam, and H. Sarbazi-Azad, *Semiconductive memory device*, US Patent App. 17/846,102, Oct. 2022.

Teaching Experience

University of Shahid Beheshti

Instructor — Advanced Computer Architecture (Fall 2024-online).

Sharif University of Technology

Teaching Assistant — Advanced Computer Architecture (2022), Interconnection Networks (Fall 2017, 2019, 2021), Hardware Description Language (Spring 2016, 2017), Numerical Computation (Spring 2015), 40-103: Computer Architecture Lab (Summer 2015),

Skills

- Programming — C/C++, Python, CUDA, BASH, Verilog; Familiar with Matlab.
- Documentation — LaTeX, Microsoft Office.
- Simulators — GPGPUSIM, Gem5, ModelSim. Cadence, Design Compiler,

Activities

- PC Member — ICS 2025, ICPE 2025
- Reviewer — IEEE Computer Architecture Letters (2022, 2023, and 2024), Sustainable Computing: Informatics and Systems (2024),

References

Patrick Eugster Professor in Computer Science and director of Software Systems (SWYSTEMS) group part of USI's Computer Systems Institute. Email: `eugstp@usi.ch`

Hamid Sarbazi-Azad Professor in the Computer Engineering Department at Sharif University of Technology and Professor in the School of Computer Science at Institute for Research in Fundamental Sciences (IPM). Email: `azad@sharif.edu`

Ahmad Khonsari Associate Professor of School of Electrical and Computer Engineering (ECE) at University of Tehran. Email: `a_khonsari@ut.ac.ir`